

Measuring AI Ability to Complete Long Software Tasks

50% Task-Completion Time Horizon Doubling Every ~7 Months

GPT-2: ~2 seconds → o3: ~1h 50m | $R^2 = 0.97$ | 207-day doubling time

Thomas Kwa, Ben West, Joel Becker et al.

Model Evaluation & Threat Research (METR) · arXiv:2503.14499 · March 2025

A New Metric: 50% Task-Completion Time Horizon

Human-grounded, intuitive measure of autonomous AI capability

Family	Length	Description
find_shell_script	3 seconds	Multiple choice: “Which file is a shell script?” Choices: “run.sh”, “run.txt”, “run.py”, “run.md”
wikipedia_research	1 minute	Research simple factual information from Wikipedia and provide accurate answers to straightforward questions.
oxdna_simple	9 minutes	Detect and fix a bug in the input files for a molecular dynamics simulation using the oxDNA package.
munge_data	56 minutes	Write a Python script to transform JSON data from one format to another by inferring the conversion rules from provided example files.
cuda_backtesting	8 hours	Speed up a Python backtesting tool for trade executions by implementing custom CUDA kernels while preserving all functionality, aiming for a 30x performance improvement.

01

Assemble Task Suite

170 diverse software & research tasks spanning seconds to 8 hours of skilled human effort

02

Measure Human & AI Performance

Time skilled humans on each task (>800 baselines, 2,529 hrs total); measure AI agent success rates

03

Fit Logistic Model

Estimate the task duration at which the AI achieves exactly 50% success — the 50% Time Horizon

Result: a single number per model — “this model can autonomously complete tasks that take skilled humans up to X minutes, 50% of the time.”

170 Tasks Spanning 3 Seconds to 8 Hours of Human Effort

>800 human baselines · 2,529 total human hours collected

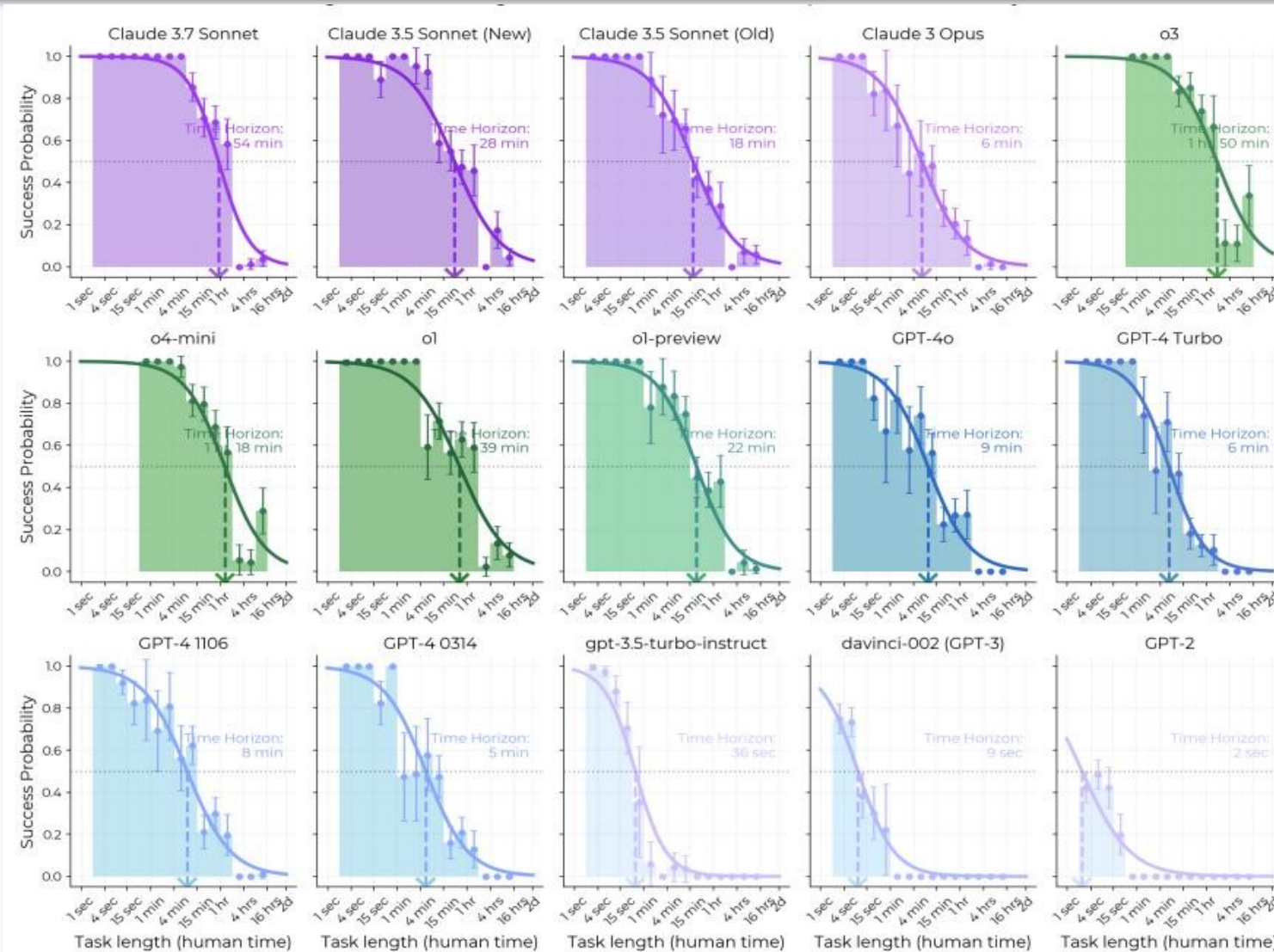
Suite	Tasks	Duration Range	Focus Area	Example Task
HCAST	97	1 min – 30 hrs	Diverse software & research	munge_data: write Python to transform JSON (56 min)
RE-Bench	7	All ~8 hrs	Hard ML research engineering	cuda_backtesting: CUDA kernels for 30× speedup (8 hrs)
SWAA	66	1 sec – 30 sec	Short single-step atomic actions	find_shell_script: identify shell script file (3 sec)

Representative Tasks (by Human Completion Time)

Task	Human Time	Description
find_shell_script	3 seconds	Multiple choice: 'Which file is a shell script?'
wikipedia_research	1 minute	Research simple factual information from Wikipedia
oxdna_simple	9 minutes	Detect and fix a bug in molecular dynamics simulation input files
munge_data	56 minutes	Write a Python script to transform JSON data by inferring conversion rules
cuda_backtesting	8 hours	Implement custom CUDA kernels for 30× speedup of a backtesting tool

AI Task Horizon: 2 Seconds (GPT-2) → ~1 Hour 50 Min (o3) in 6 Years

Exponential growth · Doubling time ≈ 207 days (~7 months) · $R^2 = 0.97$



207 days

Doubling time
(~7 months)

$R^2 = 0.97$

Trend fit across
6 years of models

GPT-2 → o3

~2 sec to ~1h 50m
(×3,300 increase)

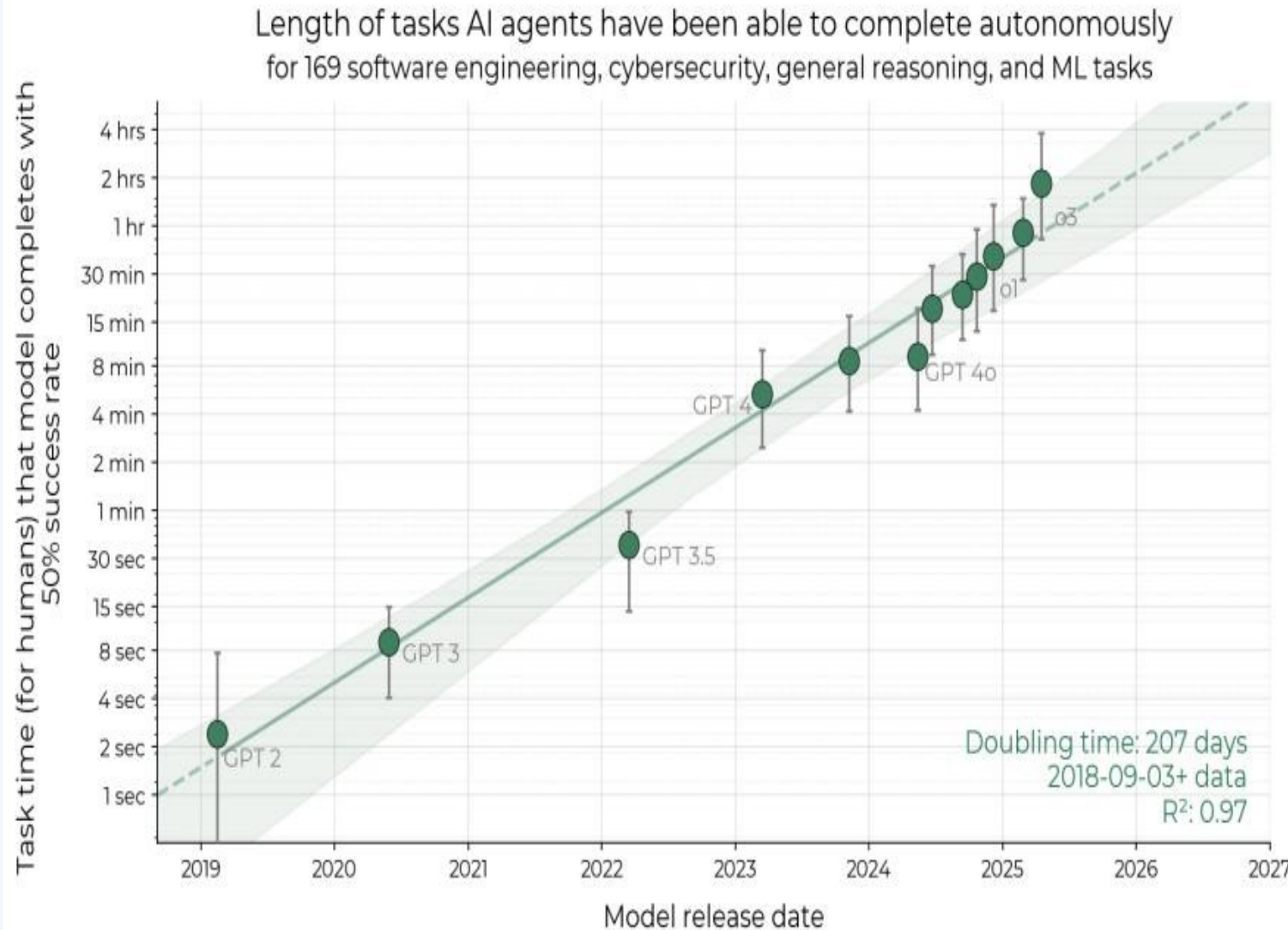
2023–2025

Key Milestones

GPT-2 (2019): ~2 sec GPT-4 (2023): ~5 min
GPT-4o (2024): ~9 min Claude 3.7: ~54 min
o3 (2025): ~1 hr 50 min

Success Probability Drops Sharply as Task Length Increases for All Models

Frontier models push the curve rightward — but steep declines persist



Reading the Curves

- **Each curve:**
shows a model's success rate vs. task duration (log scale on x-axis).
- **50% Time Horizon:**
= the task duration at which the curve crosses 50% — the core metric.
- **Rightward shift:**
= capability gain: newer frontier models succeed on longer tasks.
- **Steep drop-off:**
persists across all models — long tasks remain hard even for o3.
- **GPT-2 (2019):**
curve crosses 50% at ~2 seconds of human effort.
- **Claude 3.7 Sonnet:**
reaches 50% at ~54 minutes — a 1,600× improvement over GPT-2.
- **o3 (2025):**
pushes the frontier to ~1 hour 50 minutes (50% horizon).
- **80% horizon:**
is roughly 5× shorter than the 50% horizon for each model.

Reasoning, Tool Use, and Error Recovery Drive Time Horizon Growth

Four capability drivers identified · Key limitations noted

Improved Logical Reasoning

Models can now decompose multi-step software problems, plan sequences of actions, and maintain consistent reasoning across long task contexts.

Better Tool Use

Modern agents reliably invoke code execution, file I/O, web search, and shell commands — dramatically expanding what they can accomplish autonomously.

Reliability & Self-Awareness

Frontier models better recognise when they have made an error, can pause and re-evaluate, and are less likely to silently produce wrong outputs.

Adapting to Mistakes

When sub-tasks fail, newer models recover gracefully — retrying, reformulating, or escalating — rather than propagating errors through the full pipeline.

Key Limitations:

Performance is notably lower on less-structured, 'messier' real-world tasks. External validity is uncertain — the benchmark tasks may not perfectly represent average real-world intellectual labor distributions.

Extrapolation Predicts Month-Long Task Automation by 2028–2031

Naive trend extrapolation — subject to significant uncertainty

2023
GPT-4

~5 min

2024
Claude 3.5

~28 min

EXTRAPOLATED



~1 Month Horizon Target

167 work hours $\approx$ 1 month of software engineering.
Naive extrapolation: reached between mid-2028 and mid-2031 if current doubling continues.

2023–2025 Acceleration

Recent growth rate is ~20% faster than the 2019–2025 average, suggesting the trend may be accelerating rather than slowing.

Significant Caveats

Extrapolation assumes stable doubling time and unchanged task structure.
Real-world barriers (regulation, safety constraints, task complexity) may slow or alter trend.

Workforce Implications

Month-long autonomous task capability implies potential automation of many end-to-end software projects — with major economic and safety implications.

Takeaways: Exponential Autonomous Capability Growth Demands Proactive Safety Governance

1

Robust, Intuitive Metric

The 50% time horizon provides a human-grounded measure of AI autonomous capability, achieving $R^2 = 0.97$ across 6 years of frontier model releases.

2

Doubling Every ~7 Months — Possibly Accelerating

207-day doubling time persists from GPT-2 (2019) to o3 (2025). The 2023–2025 rate is ~20% faster, suggesting possible further acceleration.

3

Frontier Now at ~2 Hours of Autonomous Work

o3 can autonomously complete tasks that take skilled software engineers ~1h 50m — a 3,300× improvement since GPT-2 in 2019.

4

Month-Long Automation Horizon by 2028–2031

Naive trend extrapolation implies AI agents reaching a 1-month (167-hr) task horizon within the next 3–6 years, threatening widespread automation of software projects.

5

Limitations & External Validity Remain Open Questions

Lower performance on unstructured tasks and uncertain generalization to real-world job distributions mean capability estimates may not map directly onto economic impact.